

# Two ways to order the option cards

Why the study does not put the best-matching option first · VRDS Experiment 2, Block 5 · 23 September 2026

## The question

Each scenario shows six options. Something has to decide which one appears first. Two designs were considered:

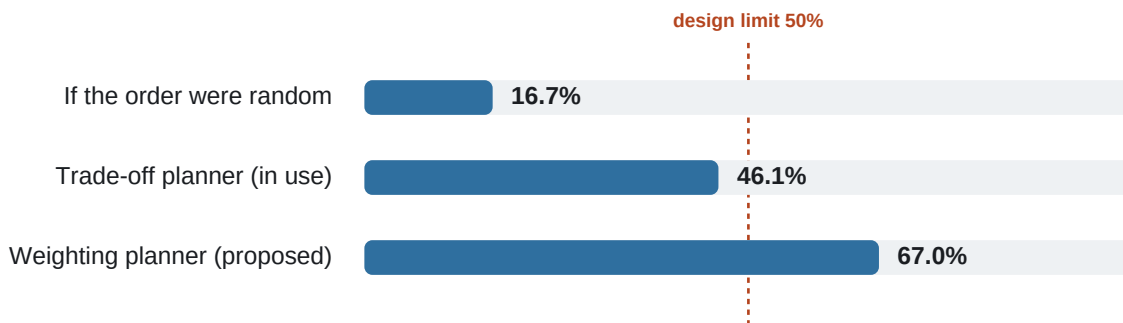
- **The weighting planner.** Give each of the person's four values a weight, score every option, and put the highest score first. The first card is then the option that best matches the person.
- **The trade-off planner (the one in use).** Compare the options two at a time and ask which value has to give way in that pair. The first card is the option that survives those comparisons, which is not always the best match.

## The short answer

The weighting planner is easier to explain, and it is the wrong choice here. It would put the best-matching option first so often that the study could no longer tell two different things apart: **people choosing what matches their values**, and **people choosing whatever is at the top of the list**.

## How often is the first card also the best-matching option?

Scenario 1, measured over 5,000 simulated participants:



Higher is worse. At 100% the first card would always be the best match, and the order alone would tell the participant which option the study thinks suits them.

## The same measurement, every scenario

5,000 simulated participants per scenario. Each one has a different ordering of the four values and different strengths behind that ordering, so the sample covers people who care about very different things. The percentages are how often the first card is also the best-matching option.

| Scenario                   | By chance    | Trade-off (in use) | Weighting    |
|----------------------------|--------------|--------------------|--------------|
| 1 · Chemical leak          | 16.7%        | 46.1%              | 67.0%        |
| 2 · Wildfire               | 16.7%        | 49.1%              | 70.5%        |
| 3 · Cancer doses           | 16.7%        | 57.6%              | 76.7%        |
| 4 · Care hours (deciding)  | 16.7%        | 43.7%              | 66.5%        |
| 5 · Care hours (receiving) | 16.7%        | 43.7%              | 66.5%        |
| <b>Average of the five</b> | <b>16.7%</b> | <b>48.0%</b>       | <b>69.4%</b> |

Simpler weights — 4, 3, 2, 1 by rank instead of the person's own scores — give 65.4%. The problem is the design, not the particular weights.

## What those numbers mean

- **16.7% is the floor.** With six options, a random order puts the best match first one time in six. No design can go below this.
- **The trade-off planner sits at 48%.** About half the time the first card is the best match, and about half the time it is not. So when somebody picks the first card, the data still contains people who did it for each reason, and the two can be told apart.
- **The weighting planner sits at 69%.** More than two times in three, the first card is the best match. Choosing the top card and choosing the best match stop being separate events.
- **The study has a written limit of 50%** for exactly this reason. The trade-off planner is inside it; the weighting planner is 19 points outside it.

## Why a high percentage damages the study

- **The main score stops being trustworthy.** The consistency score (VCI) counts how often somebody picks the option that fits their values. People are known to pick the first thing they see. If the first thing is usually the best fit, a high score can mean careful choosing or lazy clicking, and nothing in the data separates them.
- **It quietly tells the participant the answer.** The study never shows anybody which option matches their values — that is deliberate, because being told changes the remaining answers. If the best match is nearly always first, the position says it anyway.
- **It weakens every later comparison.** Position effect, stability and the prediction test all read the choice as a decision about content. If the order is doing part of the choosing, every one of those numbers carries the same doubt.

## What the trade-off planner does instead

- It uses the participant's own values, from Blocks 1 to 4 — not a guess and not a default.
- It compares options in pairs. It may set aside the top value when the gap on it is tiny and the gap on the second value is large. That is a real trade-off, and it is why the first card is not simply the best match.
- It has **no tuning numbers**. Every threshold comes from that participant's own answers, so there is nothing in it chosen to make the results look a certain way.
- It leaves performance out of the ordering on purpose. Nothing in Blocks 1 to 4 measures whether speed matters more than reversibility, so weighting them would mean inventing a preference and presenting it as the participant's own.

## What is given up by not switching

This is the honest cost. The weighting planner is one sentence to describe; the trade-off planner needs a paragraph and a worked example. Ordering is harder to explain in a methods section. That is a writing problem. The alternative is a measurement problem, and a measurement problem cannot be fixed after the data is collected.

## If the advisor prefers a simpler design

There is one alternative that is also defensible: **put the six options in a random order**, different for each participant. It removes the problem completely, because position then carries no information at all.

- **For it:** simple to describe, and the cleanest possible answer to a reviewer asking about order effects.
- **Against it:** every participant then sees an arrangement with no meaning, and the study loses the part of the design where the ordering itself reflects the person's own trade-offs. The work already done on position effects would have to be rebuilt.

## How to check these numbers

The measurement is a script in the project, not a calculation done by hand. It runs the real ordering code and the real matching code over the same simulated participants:

| Command      | npm run report:planner                |
|--------------|---------------------------------------|
| Participants | 5,000 per scenario                    |
| Random seed  | 20260923 — the same numbers every run |
| File         | tools/planner_vs_weighting.cjs        |

## One sentence for the methods section

“Option order was produced by a pairwise trade-off procedure derived from each participant's own value ranking. Order was deliberately not aligned with value fit: across 5,000 simulated participants the first-ranked option was also the best-fitting option 48% of the time, against 16.7% by chance and 69% under a weighted-sum alternative, so that choosing the first option and choosing the best-fitting option remain separable events.”

Prepared for the advisor meeting. Every percentage in this note was measured with the command above; none was estimated.